

Duration of antibody protection project

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1 Project question and aims

Consider a vaccine trial, or its long-term extension, in which an antibody marker is measured repeatedly after vaccination. Antibody levels often fall quickly at first and then more slowly. We want to estimate when the marker will fall below a level associated with protection, even if that point lies beyond the period observed so far. This creates a practical decision at each planned trial review:

Given the number and timing of the measurements collected so far, how well can we estimate the duration of antibody-associated protection, and how much would we gain from further follow-up?

The project has two core aims, followed by two extensions:

1. under a known trajectory model, determine how well its parameters and the duration of protection can be identified as the length and frequency of follow-up vary;
2. when the trajectory model is unknown, quantify the additional uncertainty from considering several plausible long-term patterns;
3. examine how differences between participants affect these results and the balance between recruiting more participants and measuring each participant more often;
4. turn the findings into practical guidance on how long, and how often, participants should be followed.

Ending follow-up for this durability question is not the same as ending all trial activity. Safety surveillance, clinical endpoints and regulatory commitments may require the trial to continue for other reasons. This project asks only when enough information has been collected to answer the durability question.

The problem is most apparent when protection is expected to last for years. The pivotal Shingrix trials followed participants for roughly three to four years, and their long-term extension ultimately assessed efficacy and immunogenicity up to 11 years after vaccination (Strezova et al. 2025). Such studies show the value of long-term follow-up. They also raise a practical question: could the antibody data available at an earlier review already have supported a reliable conclusion, or were later measurements needed to establish the long-term pattern?

Protection after COVID-19 vaccination or infection usually wanes over months rather than years. Even so, COVID-19 is a useful first case study: neutralisation predicts protection, and published studies link antibody titres with variants, boosting and vaccine effectiveness (Khoury et al. 2021;

Cromer et al. 2022; Gilbert et al. 2022; Feng et al. 2021). We can therefore develop and test the methods in a comparatively rich setting before potentially applying them to pathogens for which protection may last for years but the data, or the evidence for an immune correlate, are more limited.

2 Defining duration of protection

Let $c_i(t)$ denote the underlying level of the antibody marker for participant i at time t . The marker could be a neutralisation titre, a binding-antibody concentration or another assay measurement. Let c_{thr} denote a protection-relevant threshold on the same scale. We use the first time the marker crosses this threshold as an indication of the participant’s duration of protection:

$$T_i^* = \inf\{t \geq 0 : c_i(t) \leq c_{\text{thr}}\}. \quad (1)$$

We note that this quantity depends on how protection and the threshold are defined, and that it has no finite value if the antibody level settles above the threshold.

In the initial COVID-19 application, $c_i(t)$ and c_{thr} refer specifically to neutralising-antibody titres. Let c_{conv} be the mean neutralisation level among convalescent people, and divide titres by this reference, so that the mean convalescent level is 1. One option is to set c_{thr} to the relative titre that Khoury et al. (2021) associated with 50% protection. Alternatively, we could use their full relationship between titre and protection to convert the fitted antibody trajectory into a protection curve $E_i(t)$. The latter approach makes clear that antibody persistence and protection are related but are not the same thing. Antibody markers can provide useful quantitative predictions of protection even though they capture only part of the underlying immune response (Plotkin and Gilbert 2012; Earle et al. 2021).

Differences between participants create a further distinction. The distribution of their individual T_i^* values is not the same as the crossing time calculated from an average antibody curve. A useful population measure might instead be the time at which the proportion of people still above the threshold falls below a chosen value. A later stage of the project will examine both the individual and population questions.

3 Proposed analysis programme

The central idea is to simulate complete antibody histories for a set of participants and recreate the information that would have been available at a series of trial reviews. At each review, we would hide all later measurements, fit one or more models to the data collected by that point, and estimate T^* . Because the complete simulated histories and generating models are known, we can first isolate how much information the trial design provides and then quantify the additional uncertainty that arises when the correct model is unknown.

The workshop should first agree on the clinically relevant quantity and on what result would justify ending durability follow-up. This requires an acceptable level of uncertainty, an acceptable risk of a wrong decision, and a limit on extrapolation beyond the observed data. We should also decide whether COVID-19 is primarily a worked example or the main application.

The work could then proceed in three parts:

1. **Known trajectory model: parameter identifiability.** Begin with a biphasic model used both to generate and fit the data. Vary the review date, visit schedule, sample size, assay noise and quantification limits. At each review, assess how well the model parameters

and T^* are identified. This isolates the information provided by the amount and pattern of observation.

2. **Unknown trajectory model: model uncertainty.** Generate data from each plausible trajectory in turn, and later from one of the more detailed biological models. Fit the full set of candidate models without assuming which is correct. At each review, separate the uncertainty within each fitted model from the uncertainty arising from the choice of model, and assess how both affect estimation of T^* .
3. **Extensions and practical interpretation.** Allow participants to differ in their peak antibody levels and rates of decline; compare individual and population outcomes; and compare the value of measuring more participants with that of measuring each participant more often. If useful, convert antibody trajectories into protection against symptomatic and severe disease, and apply the same workflow to a suitable open dataset.

The main output would be a set of conditions under which estimates of duration are reliable enough to inform follow-up decisions. These conditions would distinguish limitations caused by insufficient data under a known model from those caused by uncertainty about the model itself.

4 Models of antibody decline

The analysis requires two kinds of model. Relatively simple models will be fitted to the measurements available at each trial review. More detailed biological models can then generate test data with features that the fitted models may not capture. Keeping these roles separate makes it possible to ask whether a simple analysis remains useful when its assumptions are not exactly true.

Models to fit

Each fitted model makes a different assumption about what happens after follow-up ends. We suggest using the biphasic exponential as the reference for the first analyses, while keeping the other models as credible alternatives.

Single exponential

$$c(t) = c_0 e^{-kt}. \quad (2)$$

This is the simplest model. The log concentration falls at a constant rate, and $T^* = k^{-1} \log(c_0/c_{\text{thr}})$ when $c_0 > c_{\text{thr}}$. If antibody levels really do follow this pattern, relatively short follow-up may be enough to estimate the curve. If they do not, the model wrongly extends the initial rate of decline indefinitely. Describing COVID-19 neutralisation in terms of a single half-life is nevertheless common (Khoury et al. 2021; Cromer et al. 2022).

Biphasic exponential (suggested default)

Biphasic models describe an initial rapid fall followed by a slower, more persistent decline, and have been used successfully to model antibody responses after vaccination (M. T. White et al. 2015; Hogan et al. 2023). We suggest this family as the initial reference because its parameters have useful biological interpretations, the part that governs long-term behaviour is explicit, and published COVID-19 estimates are available for calibration. It is a practical basis for comparison,

not a claim that biphasic decay will suit every dataset. Following Hogan et al. (2023), we write the model as

$$c(t) = c_0 \frac{\exp\left[-(\log 2) \left(\frac{t}{h_1} + \frac{t_s}{h_2}\right)\right] + \exp\left[-(\log 2) \left(\frac{t}{h_2} + \frac{t_s}{h_1}\right)\right]}{\exp[-(\log 2)t_s/h_1] + \exp[-(\log 2)t_s/h_2]}, \quad h_1 < h_2. \quad (3)$$

The fast component can represent the loss of short-lived antibody-secreting cells, and the slow component a longer-lived cell population. This parameterisation ensures that $c(0) = c_0$ and that the two components contribute equally at the switching time t_s . The slow half-life h_2 determines the long-term behaviour, but may be hard to estimate until the fast component has subsided.

Power-law decay

$$c(t) = c_0(1 + \alpha t)^{-\beta}. \quad (4)$$

Here the rate of decline slows continuously. Over a short period the curve can look exponential, yet it predicts that antibodies will persist much longer. Long-term antibody measurements give biological grounds for considering this pattern rather than assuming a constant half-life (Amanna et al. 2007).

Plateau or set-point

$$c(t) = A_\infty + (c_0 - A_\infty)e^{-kt}. \quad (5)$$

In this model, the antibody level settles at a non-zero value. It crosses the threshold only if $A_\infty < c_{\text{thr}}$. With short follow-up, a true plateau may be indistinguishable from a very slow decline. Long-term measurements showing persistent antibody levels motivate this option (Amanna et al. 2007). Because T^* may not have a finite value under this model, the workshop should decide whether and how to include it in model comparisons.

More complex models for generating test data

Mechanistic antibody-production model

A more detailed model can represent short- and long-lived antibody-secreting cells, as well as the clearance of the antibodies themselves:

$$\frac{dP_s}{dt} = -\delta_s P_s, \quad \frac{dP_l}{dt} = -\delta_l P_l, \quad \frac{dc}{dt} = \rho_s P_s + \rho_l P_l - \delta_c c. \quad (6)$$

When antibodies clear more quickly than the cells, this model behaves like biphasic decay. Adding immune memory or cell replenishment can produce still slower long-term decline. Related models have been developed for plasma-cell dynamics and vaccine responses (Andraud et al. 2012; Pasin et al. 2019). For this project, such a model may be most useful for generating realistic test data, rather than as the default model fitted to every dataset.

COVID-19 semi-mechanistic model

A second option for generating test data is the semi-mechanistic model developed in a series of COVID-19 studies (Nishiyama et al. 2023; Nakamura et al. 2024; Park et al. 2025; Hart et al. 2025). It links a waning mRNA stimulus to delayed, non-linear antibody production and antibody loss, and can account for boosting, infection history and differences between participants. Once the vaccine stimulus has disappeared, however, the model approaches a single exponential decline.

5 Measurement error and starting values for the biphasic model

Assay results are noisy measurements of an underlying positive trajectory. A simple starting point is to assume multiplicative log-normal error:

$$\log y_{ij} = \log c_i(t_{ij}) + \epsilon_{ij}, \quad \epsilon_{ij} \sim N(0, \sigma_{\log}^2). \quad (7)$$

Results below or above the assay’s quantification limits can be treated as censored, rather than assigned exactly to the relevant limit. This observation model can be combined with any of the trajectories in Section 4.

Table 1 gives possible starting parameters for a biphasic exponential model of BNT162b2 after dose 3 against the Delta variant, based on the parameterisation of Hogan et al. (2023). All antibody levels are expressed relative to the mean convalescent neutralisation level defined above.

Table 1: Tentative starting values for a BNT162b2 dose-3 biphasic COVID-19 neutralising-antibody baseline.

Quantity	Starting value	Source / basis	Interpretation / possible variation
Early level c_0	0.92	Hogan et al. (2023)	BNT162b2 dose 3 against Delta; 95% CrI 0.71–1.19
Log-error SD $\sigma_{\log}$	0.20	Workshop noise choice	Explore lower and higher assay variability
Fast half-life h_1	35 days	Hogan et al. (2023)	Early contraction; tentative range 25–60 days
Slow half-life h_2	581 days	Hogan et al. (2023)	Most important long-term assumption; consider 250–900 days
Switching time t_s	75 days	Hogan et al. (2023)	Equal contributions from the two components at this time
Protection anchor c_{thr}	0.202 / 0.030	Khoury et al. (2021)	Relative titres associated with 50% symptomatic / severe protection

6 Differences between participants

In the biphasic reference model, participants can differ in their peak antibody level and slow half-life. One way to represent these differences is with correlated log-normal random effects directly on those two quantities:

$$c_{0i} = c_0 e^{\eta_{0i}}, \quad h_{2i} = h_2 e^{\eta_{hi}}, \quad (8)$$

$$\boldsymbol{\eta}_i = \begin{pmatrix} \eta_{0i} \\ \eta_{hi} \end{pmatrix} \sim N_2(\mathbf{0}, \boldsymbol{\Sigma}), \quad \boldsymbol{\Sigma} = \begin{pmatrix} \omega_0^2 & \rho_{0h}\omega_0\omega_h \\ \rho_{0h}\omega_0\omega_h & \omega_h^2 \end{pmatrix}. \quad (9)$$

This structure lets us estimate outcomes for both individuals and the population. It also lets us ask whether recruiting more participants can make up for taking only a few measurements from each person. Similar structures have been used in population models of vaccine responses (Pasin et al. 2019).

Table 2 gives tentative starting values for this between-participant variation, using the same BNT162b2 dose-3 setting as Table 1.

Table 2: Tentative starting values for between-participant heterogeneity in the biphasic baseline.

Quantity	Starting value	Source / basis	Interpretation / possible variation
Peak / half-life log-SD ω_0/ω_h	0.30 / 0.25	Workshop starting value	Moderate differences between participants
Peak-half-life correlation ρ_{0h}	0.30	Workshop correlation choice	Equivalent to a -0.30 peak-rate correlation; include zero as a comparison

The uncited values in Tables 1 and 2 are convenient starting points for simulation, not estimates from the literature. The workshop may instead choose a different vaccine, dose, assay scale or long-term rate of decline.

7 Evidence base and key studies

The papers below are grouped by the part of the project they inform: interpreting antibody markers, calibrating the COVID-19 example, representing differences between participants, and understanding long-term antibody dynamics and follow-up.

From antibody markers to protection

Plotkin and Gilbert (2012). This framework distinguishes a statistical correlate, a mechanistic correlate and a surrogate endpoint. It helps us avoid treating a protection-relevant antibody marker as the whole immune mechanism (Plotkin and Gilbert 2012).

Earle et al. (2021). This comparison across vaccines gives broader evidence that antibody levels can act as a correlate of protection for COVID-19 vaccines. It supports our use of an antibody marker without implying that antibodies alone cause protection (Earle et al. 2021).

Gilbert et al. (2022). This analysis of the mRNA-1273 efficacy trial shows that neutralisation is a graded correlate of risk. It therefore supports modelling the marker continuously rather than using only a binary laboratory cut-off (Gilbert et al. 2022).

Feng et al. (2021). This analysis of ChAdOx1 nCoV-19 also finds a continuous relationship between immune markers and symptomatic or asymptomatic infection. It cautions against treating one threshold as universal (Feng et al. 2021).

Khoury et al. (2021). This study provides the main quantitative link between neutralisation titre and protection. Its logistic relationship and estimates of the titres associated with 50% protection can be used to convert a fitted antibody trajectory into a protection curve for a specific clinical endpoint (Khoury et al. 2021).

Cromer et al. (2022). This meta-analysis extends the link between neutralisation titre and protection across variants and boosting. It shows that the initial titre and antigenic change can alter the duration of protection even when antibodies wane at the same rate (Cromer et al. 2022).

COVID-19 antibody kinetics and calibration

Hogan et al. (2023). This modelling study estimates initial neutralising-antibody levels for specific vaccines and doses, together with fast and slow half-lives and a switching time. These estimates provide the tentative COVID-19 calibration and show how antibody dynamics can be linked to longer-term effectiveness against variants (Hogan et al. 2023).

Nishiyama et al. (2023). This study extends an antibody model based on mRNA stimulation to include a booster dose and previous infection. It shows how repeated exposures can alter the duration of the response and provides a compact semi-mechanistic model for later simulations (Nishiyama et al. 2023).

Variation between participants

Nakamura et al. (2024). This study combines frequent measurements from a small group with sparse measurements from more than 2,400 participants to reconstruct individual trajectories after primary vaccination. Its fitted parameter distributions are useful for calibrating variation between participants, although the clinical data are access-restricted (Nakamura et al. 2024).

Park et al. (2025). This study groups antibody trajectories after primary and booster vaccination into durable, vulnerable and rapidly declining patterns. The link between rapid decline and earlier subsequent infection shows that differences between participants have clinical importance (Park et al. 2025).

Hart et al. (2025). This study links the same individual antibody model to susceptibility and population outbreak risk. Its open code and derived parameter estimates can be used directly for simulation and calibration (Hart et al. 2025).

Pasin et al. (2019). This Ebola prime–boost study shows how biological antibody dynamics and differences between participants can be estimated together. It is therefore a useful guide for the stage of the project that introduces participant variation (Pasin et al. 2019).

Long-term dynamics and follow-up

Amanna et al. (2007). Long-term measurements across antigens show that antibody persistence varies greatly and need not follow one constant half-life. This finding motivates the power-law and plateau models, and the project’s focus on long-term behaviour (Amanna et al. 2007).

Andraud et al. (2012). These linked plasma-cell and antibody models provide a clear biological basis for generating trajectories over several time scales. They can be used to test when a simpler decay curve is a good enough approximation (Andraud et al. 2012).

White et al. (2015). This analysis of the RTS,S malaria vaccine is the closest published example of the proposed approach: it uses a biphasic model of antibody decline and links that decline to the duration of vaccine efficacy (M. T. White et al. 2015).

Strezova et al. (2025). The final Shingrix analysis shows the practical scale of this design problem. The extension of the pivotal phase 3 trials followed more than 7,000 participants and assessed protection for up to 11 years. It provides a concrete example in which the value of more durability data must be weighed against the burden of prolonged follow-up (Strezova et al. 2025).

Selected literature

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